Task2 Machine Learning Model

In [1]:
```
import pandas as pd
```

In [8]:
```
df = pd.read_excel("Dataset.xlsx")
```

In [7]:
```
import os
os.chdir("E:\\Shraddha\\Project")
```

In [9]:
```
df.head()
```

Out[9]:

	job_id	job_title	job_category	experience_level	years_of_experience	education_required	annual_salary_usd	salary_min_
0	AIJOB0001	AI Agent Developer	AI Engineering	Senior (6-9 yrs)	7	Master's	239000	155
1	AIJOB0002	Prompt Engineer	AI Engineering	Senior (6-9 yrs)	2	Bachelor's	166000	90
2	AIJOB0003	LLM Engineer	AI Engineering	Senior (6-9 yrs)	4	Associate's	360000	160
3	AIJOB0004	Data Engineer (AI)	Data Engineering	Senior (6-9 yrs)	3	Bachelor's	161000	130
4	AIJOB0005	AI Product Manager	Product	Lead (10+ yrs)	5	Bootcamp/Self-taught	283000	140

5 rows × 25 columns

In [11]:
```
df = df[['experience_level', 'job_category', 'annual_salary_usd']]
df.head()
```

Out[11]:

	experience_level	job_category	annual_salary_usd
0	Senior (6-9 yrs)	AI Engineering	239000
1	Senior (6-9 yrs)	AI Engineering	166000
2	Senior (6-9 yrs)	AI Engineering	360000
3	Senior (6-9 yrs)	Data Engineering	161000
4	Lead (10+ yrs)	Product	283000

In [13]:
```
df = pd.get_dummies(df, drop_first=True)
df.head()
```

Out[13]:

	annual_salary_usd	experience_level_Lead (10+ yrs)	experience_level_Mid (3-5 yrs)	experience_level_Senior (6-9 yrs)	job_category_Architecture	job_catego
0	239000	False	False	True	False	
1	166000	False	False	True	False	
2	360000	False	False	True	False	
3	161000	False	False	True	False	
4	283000	True	False	False	False	

In [31]:
```
x = df.drop('annual_salary_usd', axis=1)
y = df['annual_salary_usd']
```

In [32]:
```
rf_model = RandomForestRegressor(n_estimators=100, max_depth=10)
```

In [33]:
```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

In [34]:
```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
```

```
model.fit(x_train, y_train)
```

Out[34]:

▾ LinearRegression

▶ Parameters

In [35]:
```
y_pred = model.predict(x_test)
```

In [36]:
```
from sklearn.metrics import r2_score
score = r2_score(y_test, y_pred)
print(score)
```

```
0.3195566131030503
```

In [37]:
```
from sklearn.tree import DecisionTreeRegressor

dt_model = DecisionTreeRegressor()
dt_model.fit(x_train, y_train)

dt_pred = dt_model.predict(x_test)

from sklearn.metrics import r2_score
print("Decision Tree Score:", r2_score(y_test, dt_pred))
```

```
Decision Tree Score: 0.3028562884813346
```

In [38]:
```
from sklearn.ensemble import RandomForestRegressor

rf_model = RandomForestRegressor()
rf_model.fit(x_train, y_train)

rf_pred = rf_model.predict(x_test)

print("Random Forest Score:", r2_score(y_test, rf_pred))
```

```
Random Forest Score: 0.30110163252173294
```

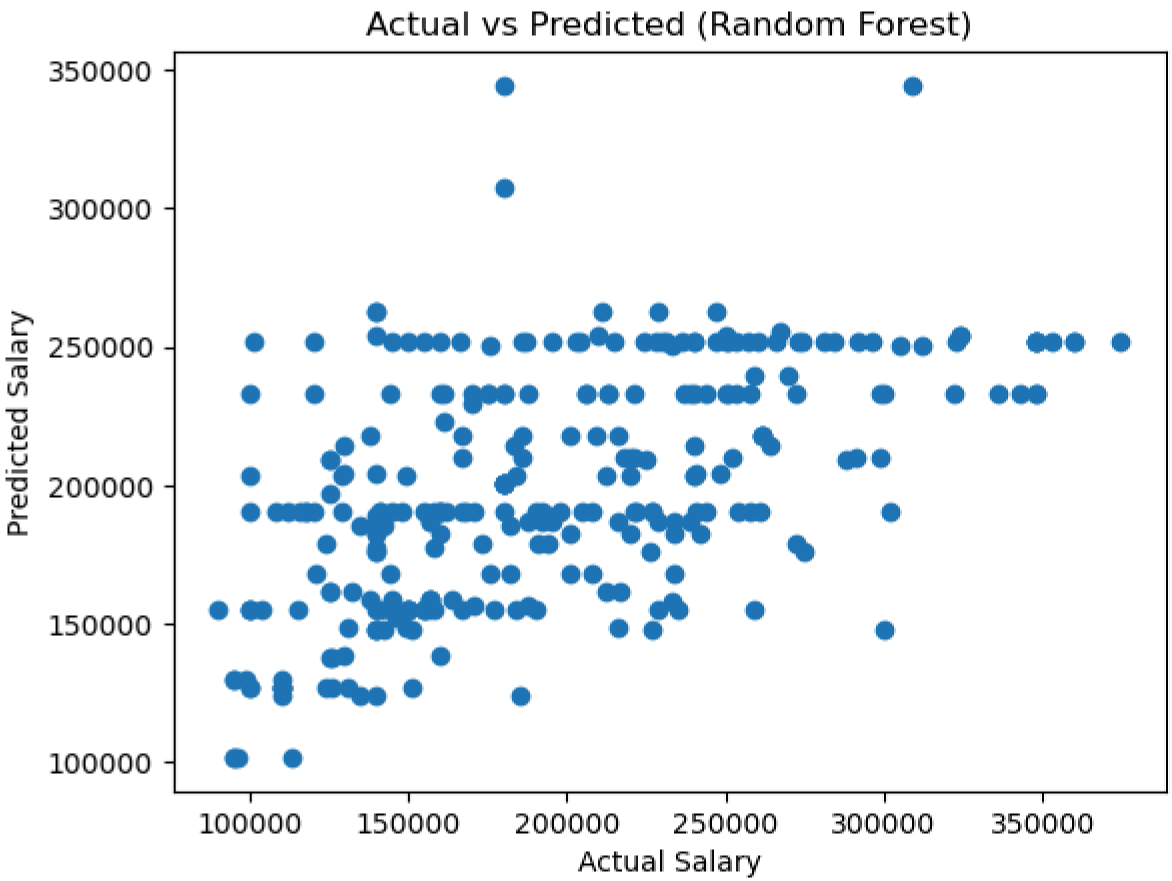
In [39]:
```
print("Linear Regression:", r2_score(y_test, y_pred))
print("Decision Tree:", r2_score(y_test, dt_pred))
print("Random Forest:", r2_score(y_test, rf_pred))
```

```
Linear Regression: 0.3195566131030503
Decision Tree: 0.3028562884813346
Random Forest: 0.30110163252173294
```

In [40]:

```python
import matplotlib.pyplot as plt

plt.scatter(y_test, rf_pred)
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.title("Actual vs Predicted (Random Forest)")
plt.show()
```

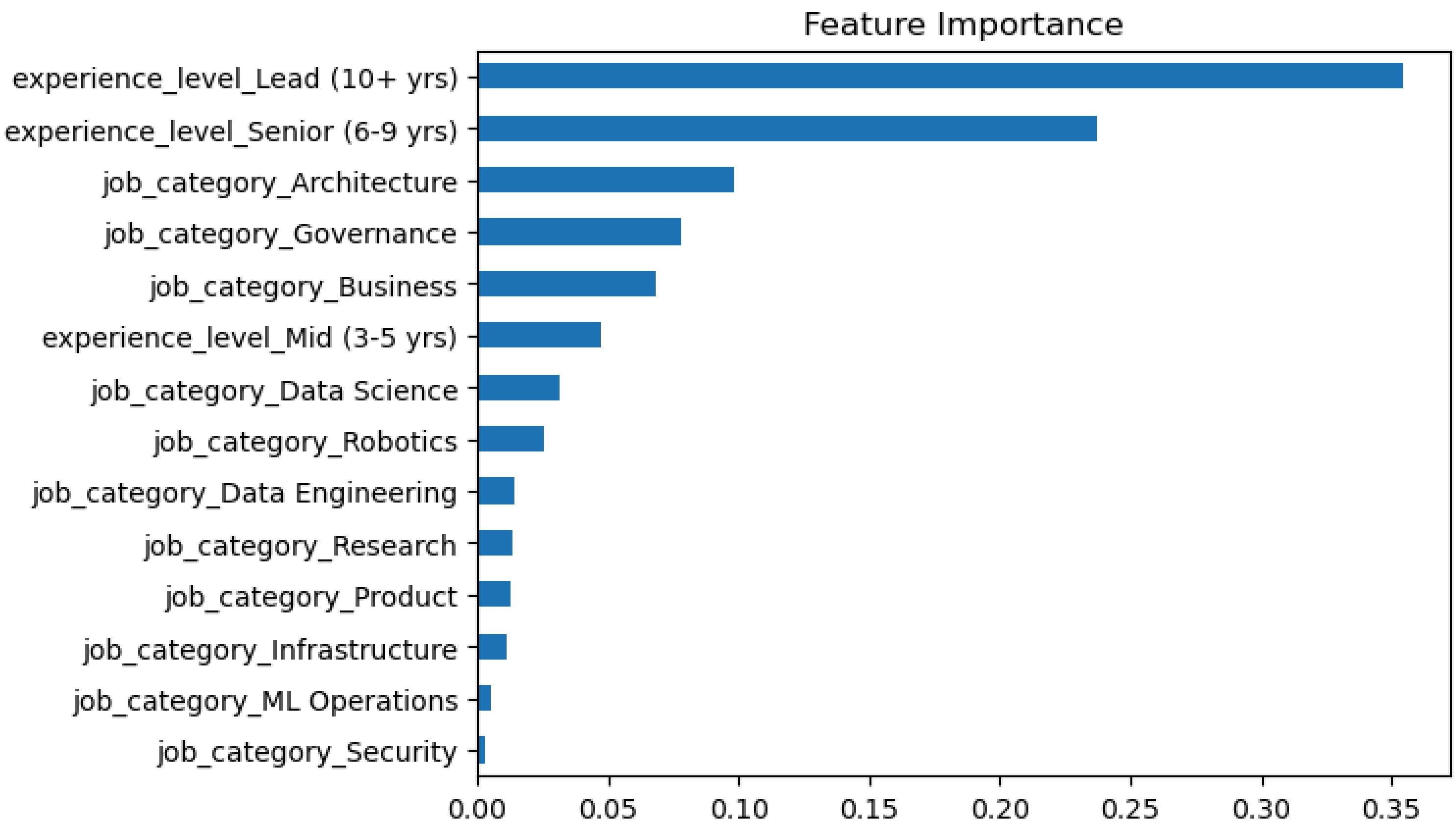


In [41]:

```python
import pandas as pd
```

```
importance = rf_model.feature_importances_
features = x.columns

feat_imp = pd.Series(importance, index=features)
feat_imp.sort_values().plot(kind='barh')
plt.title("Feature Importance")
plt.show()
```



In [ ]:

In [ ]: